



MAKE A BUZZER ALARM! THE MIDNIGHT MANGO THIEF



THE GREAT MANGO ORCHARD HEIST

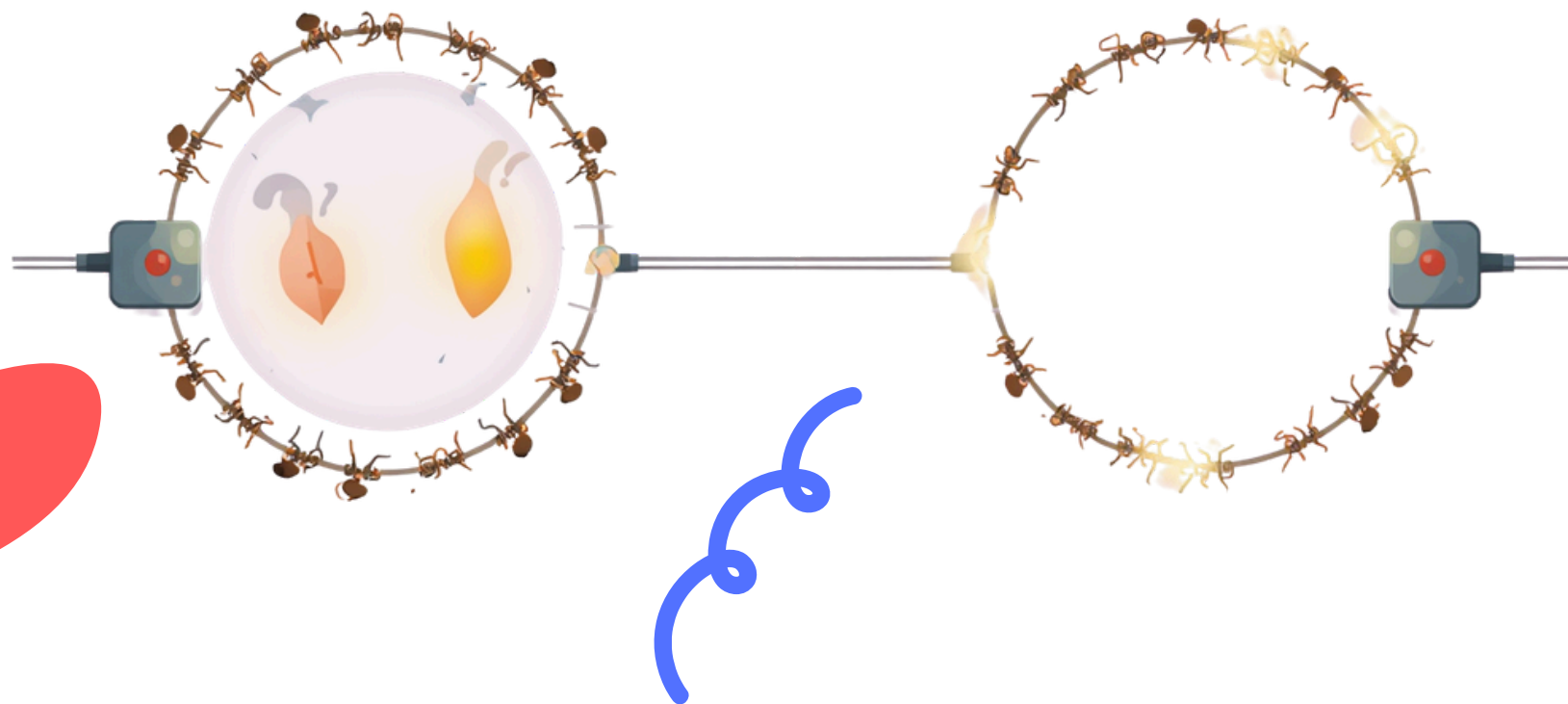
Story Mission

Ayaan's Alphonso mango is being nibbled every night! Can you help build a secret alarm to catch the thief?

Help Ayaan protect his mangoes!

WHAT IS A CIRCUIT?

Imagine tiny "Electricity Ants" marching through a wire path! They need a complete loop to deliver power and make things work.



Open Circuit: The ants cannot jump across a broken wire — no sound!

Closed Circuit: The ants march all the way to the buzzer and back — buzzer sings!

THE METRO & RICE LIGHTS EXAMPLE

Let's understand circuits with fun examples from India! When electricity has a complete path, everything works perfectly.

Metro Rail: Full circle track means train moves nonstop. Broken track stops the train!

Diwali Lights: One cut wire makes all the rice lights go off!



MEET THE SOUND MAKER: THE BUZZER!

The buzzer is a magical part that turns the electricity ants' march into sound! When the ants complete their journey through the circuit, the buzzer sings!

Doorbell goes 'Ding-Dong' when someone visits!

School period bell rings to tell you it's time for the next class!

Microwave beeps when your food is ready to eat!



THE PRESSURE SWITCH: YOUR ALARM'S TRIGGER

The pressure switch is the secret trigger for your alarm! When you push the pressure plate down, the electricity ants can finally march through and make the buzzer sing!

Push the plate, ants can march through!

Fridge door buzzer beeps when door opens too long.

Welcome doormat bells ring when guests step on them!



STEP 1 - CONNECT BATTERY TO BUZZER

Drag the red wire from the 9V battery to the buzzer's red terminal. This connects the power source to your alarm! Watch carefully as the electricity ants get ready to march through your circuit.

**Wah! You connected the
power!**



STEP 2 - ADD THE PRESSURE PLATE SWITCH

Bridge the black ground wire through the pressure plate switch. Look at the switch module and connect the wires carefully. The black wire goes from the buzzer, through the switch, and back to the battery. When someone steps on the plate, the circuit will close!

Super Engineer! The loop is almost complete!



STEP 3 - TEST THE ALARM

Drop the virtual stone on the pressure plate to close the circuit. Watch carefully as the stone presses down on the mat. When the circuit closes, the electricity ants can finally complete their march all the way to the buzzer!

Shabash! The alarm is ringing! BZZZZ!

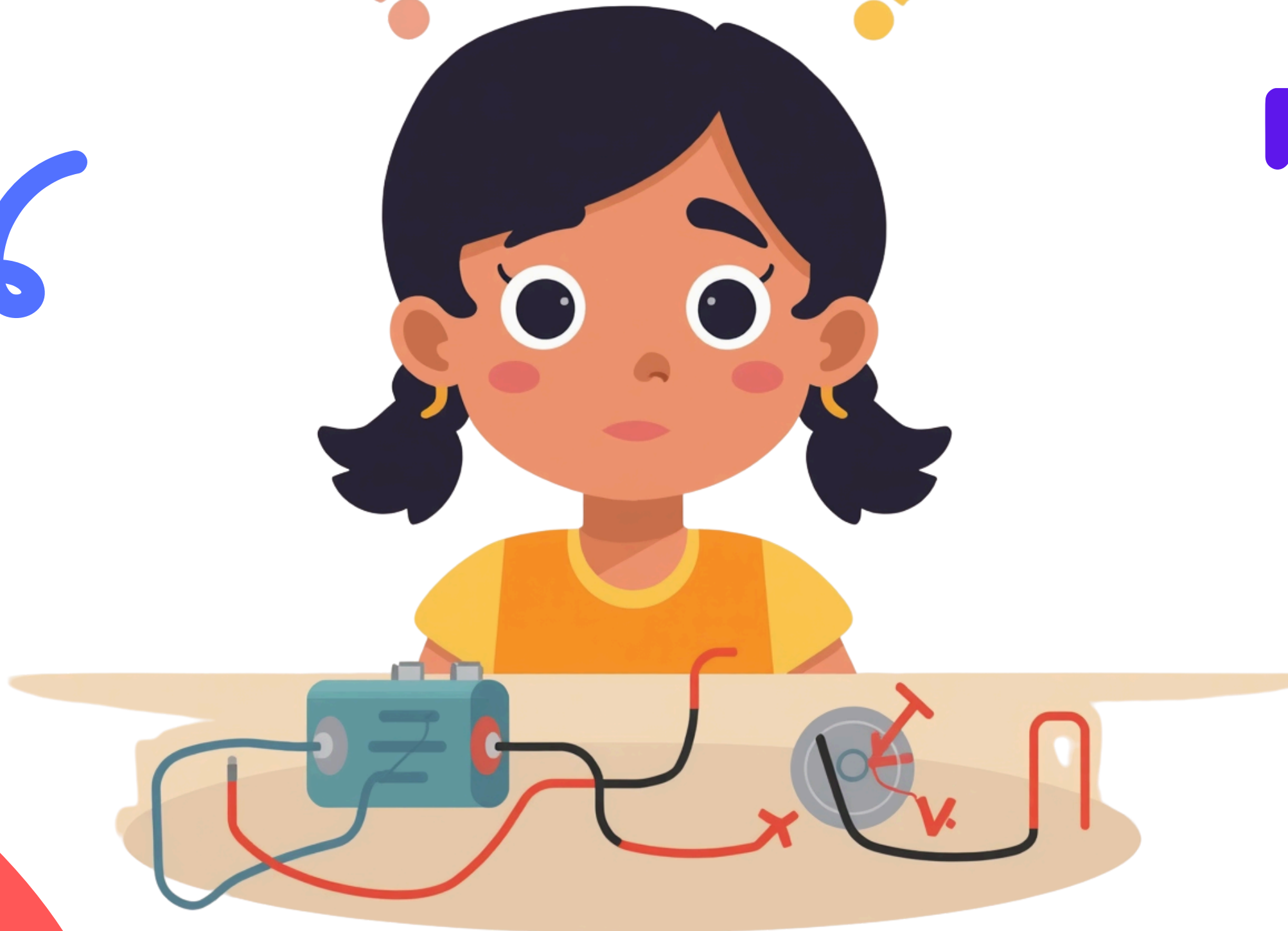


PUZZLE 1 - FIX THE WIRE POLARITY

Oops! The Wires Are Upside Down!

Oh no! The buzzer isn't ringing because the wires are connected the wrong way. Can you swap the red and black wires to fix the circuit?

Let's help the ants march the right way!





PUZZLE 2: REPLACE THE PLASTIC BLOCK

Spot the Problem!

Oh no! There's a plastic block in the circuit and the buzzer won't ring! Plastic stops the electricity ants from marching. Find the metal pin to fix it!

**Hint: Conductive means
ants can walk through!**

PUZZLE 3 - SAFE OR NOT SAFE?

Which power source is safe for young inventors like you? Look carefully at each picture and choose wisely!

Toy AA battery pack – Safe for kids!

Wall socket – Never touch! Very dangerous!

Battery with warning sign – Ask an adult first!



STEM SAFETY SCIENTIST OATH

I promise to only use toy batteries and never touch big wall plugs. I will always ask a grown-up before doing any electricity experiments!

**WAH! YOU ARE A
SAFETY SUPERSTAR!**



YOUR HOMEWORK: FIND BUZZERS AT HOME

Keep Exploring, Super Engineer!

Look for doorbells, fridge alarms, or school bells at home. Can you spot the buzzer?

Doorbell: Press and hear the ding-dong sound!

Fridge Alarm: Listen when the door stays open too long.

School Bell: The buzzer that ends your classes!



THANK YOU & KEEP INVENTING!

The Midnight Mango Thief Adventure

Thank you for joining our buzzer alarm building journey! You are now a Super Engineer who can catch mango thieves!



**Questions? Let's chat
and build more fun
circuits!**

